

Lunar Surface Analysis via XRF

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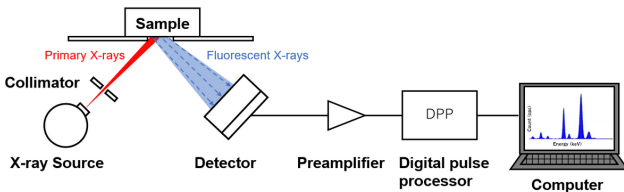
Alma Mater Studiorum - University of Bologna

May 11, 2026

- **Goal:** Identification and quantification of chemical elements on the lunar surface.
- **Methodology:** Using Machine Learning to automate and refine the interpretation of these spectra.
- **Data Strategy:** Development of a custom dataset and utilization of multiel-spectra for high-fidelity synthetic data generation.

XRF Spectroscopy Fundamentals

- High-energy X-rays/Gamma rays irradiate the sample.
- Secondary X-rays (fluorescence) are emitted, different for each element.
- The detector captures a spectrum where energy peaks correspond to specific atoms.



The analysis focuses on determining the chemical elements within the XRF spectra through three interconnected computational tasks:

- 1 Classification
- 2 Regression
- 3 Denoising

To train models, we developed a Generator that simulates physical variations in XRF spectra.

- **Composition:** Samples contain **0 to 5** elements randomly selected from **41** possible candidates.
- **Stochasticity:** To ensure model robustness, each spectrum is generated by randomly varying operating conditions and noise levels simulating real-world measurement variability.

The classification task focus on identifying the elemental composition by analyzing spectral signatures.

Analytical stages

- **Multi-label classification:** across a pool of 41 potential elements.
- **Multi-class classification:** across 5 classes (1 to 5 elements) defined by the number of elements present in the sample.

Preprocessing Strategies and Architectures

Preprocessing Scaling

Tested to handle high dynamic range in spectral counts:

- **Standard:** 0 mean and 1 variance.
- **MinMax:** Rescaling to $[0, 1]$ range.
- **Log-MinMax:** Logarithmic transformation to compress peaks.

Model Architectures

Machine Learning:

- Random Forest, AdaBoost, XGBoost

Deep Learning:

- MLP, CNN, Transformer

Results: Example Multi-label Classification

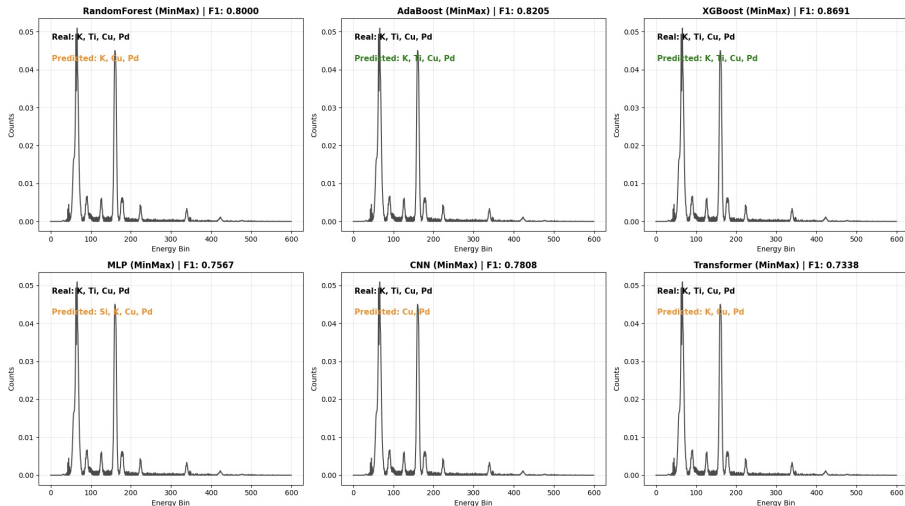


Figure: Predicted vs. real elemental composition for a sample

Results Multi-label Classification

Preprocessing	Architecture	F1-Score	Precision	Recall
Standard	XGBoost	0.8691	0.9593	0.8237
MinMax	XGBoost	0.8691	0.9589	0.8243
LogMinMax	XGBoost	0.8691	0.9589	0.8243
MinMax	AdaBoost	0.8205	0.9151	0.7821
LogMinMax	AdaBoost	0.8203	0.9088	0.7830
Standard	AdaBoost	0.8142	0.9049	0.7770
LogMinMax	RandomForest	0.8004	0.9618	0.7314
MinMax	RandomForest	0.8000	0.9600	0.7305
Standard	RandomForest	0.7995	0.9848	0.7292
LogMinMax	CNN	0.7979	0.9719	0.7267
Standard	CNN	0.7910	0.9597	0.7203
MinMax	CNN	0.7808	0.9567	0.7062
MinMax	MLP	0.7567	0.9419	0.6802
LogMinMax	MLP	0.7542	0.9494	0.6751
MinMax	Transformer	0.7338	0.9441	0.6540
Standard	MLP	0.7320	0.9351	0.6415
Standard	Transformer	0.7173	0.8914	0.6331
LogMinMax	Transformer	0.7141	0.9299	0.6295

Figure: Multi-label classification results ranked by F1-Score.

Results Multi-class Classification

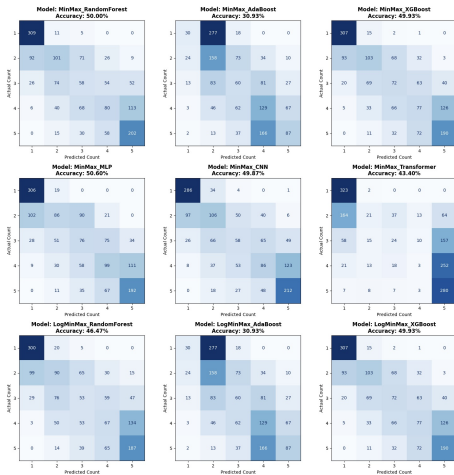


Figure: Confusion matrices for multi-class classification.

Results: Example Multi-class Classification

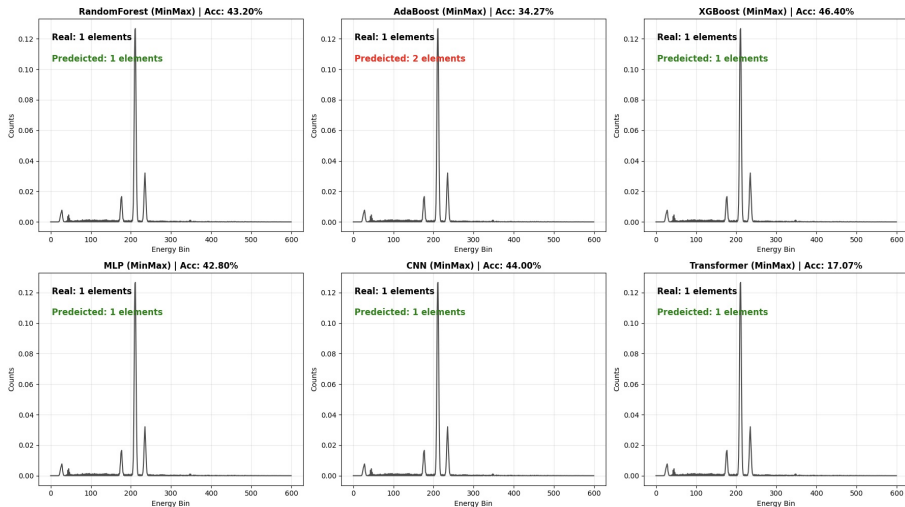


Figure: Predicted vs. real number of elements.

Results Multi-class Classification

Preprocessing	Architecture	F1-Score	Precision	Recall
LogMinMax	XGBoost	0.4473	0.4411	0.4653
MinMax	XGBoost	0.4467	0.4402	0.4640
Standard	RandomForest	0.4288	0.4313	0.4440
Standard	XGBoost	0.4240	0.4203	0.4413
MinMax	CNN	0.4207	0.4330	0.4400
LogMinMax	CNN	0.4185	0.4230	0.4307
MinMax	RandomForest	0.4137	0.4148	0.4320
MinMax	MLP	0.4023	0.4007	0.4280
LogMinMax	RandomForest	0.3820	0.3780	0.4027
LogMinMax	MLP	0.3686	0.3786	0.4133
Standard	CNN	0.3650	0.3579	0.3853
Standard	AdaBoost	0.3327	0.3489	0.3427
MinMax	AdaBoost	0.3327	0.3489	0.3427
LogMinMax	AdaBoost	0.3327	0.3489	0.3427
Standard	MLP	0.2388	0.2694	0.3013
Standard	Transformer	0.0802	0.0490	0.2213
LogMinMax	Transformer	0.0776	0.0472	0.2173
MinMax	Transformer	0.0498	0.0291	0.1707

Figure: Multi-class classification results ranked by F1-Score.

The regression task focuses on **quantifying** the elemental composition by predicting concentration values from spectral data.

Problem formulation

- **Input:** XRF spectrum — 600 energy bins (0–30 keV)
- **Output:** 41 elemental concentrations, non-negative and summing to 1
- **Constraint:** outputs lie on the probability simplex $\Rightarrow$ enforced via ReLU normalisation at the output layer

Deep Learning

- **CNNRegressor:** $3 \times$ Conv1D blocks with batch norm, global average pooling, and dense head
- **ResNetRegressor:** residual connections for better gradient flow on overlapping spectral peaks
- **TwoStageRegressor:** classifier mask applied before regression — detect then quantify

Classical Baselines

- **PLS** (Partial Least Squares): 20 latent components, standard chemometrics approach, handles multicollinearity
- **Ridge:** L2-regularised linear regression, one model per element (MultiOutputRegressor)

Both baselines use
StandardScaler + clip-to-positive
+ renormalisation.

Loss Functions & Training

Loss Functions

- **Masked MSE:** mean squared error computed only on *active* elements (ground truth > 0)
- **KL Divergence:** penalises distributional mismatch between predicted and true concentrations
- **Combined:**
 $1.0 \cdot \text{MSE} + 0.5 \cdot \text{KL}$

Training Setup

- Optimizer: Adam, $\text{lr} = 10^{-3}$
- Up to 60 epochs with early stopping (patience 10)
- Batch size: 64
- Input normalisation: StandardScaler
- Split: 2000 / 400 / 400

Pipeline Comparison

To evaluate the benefit of denoising as a preprocessing step:

Raw Pipeline

Noisy spectrum $\rightarrow$ regressor trained on noisy data

Denoised Pipeline

Noisy spectrum $\rightarrow$ CNN Autoencoder denoiser $\rightarrow$ regressor trained on clean data

- Test set: `fast_scan` config — simulates handheld XRF conditions (high noise)
- Hypothesis: denoising improves quantification accuracy, especially at high noise levels
- Evaluation metric: **Masked MAE** on active elements

Results: Regression

[PLACEHOLDER — regression results figure]

Figure: Masked MAE comparison across models and pipelines.

- **The Problem:** XRF spectra are often corrupted by noise and background artifacts.
- **The Impact:** Noise can obscure low-intensity peaks, leading to incorrect elemental identification or inaccurate quantification.
- **The Objective:** "Chemical signal recovery": mapping a noisy raw input spectrum to its clean, noise-free ground-truth counterpart.

Denoising Dataset Generator

To synthesize the required (X, Y) training pairs, we developed a high-performance generator:

- **Decisional Logic (n_counts):** Every pair (X, Y) is generated by fixing a specific physical state θ and synthesizing two versions via the `multiel_spectra` engine:
 - **Ground Truth (Y):** $n_counts = 1$ (optimal statistics/zero noise).
 - **Noisy Observation (X):** $n_counts \in [20, 100]$ (stochastic degradation).
- **Generic Learning:** This variety prevents overfitting to a fixed noise profile, forcing the model to learn pure signal recovery.

XRF spectra present a unique "**Peak vs. Baseline**" challenge: peaks can be orders of magnitude larger than the background.

- **Standard Scaling:** Z-score normalization; preserves distribution but remains sensitive to high-intensity outliers.
- **MinMax Scaling:** Normalizes to $[0,1]$; often suppresses trace signals in the presence of dominant peaks.
- **LogMinMaxScaler (Optimal):** Applies $\log(1 + x)$ followed by MinMax scaling.
 - **Benefit:** Compresses dynamic range, ensuring low-intensity peaks are not "drowned out" and remain visible to the model.

Detailed Model Architectures (1/2)

We evaluated 8 diverse architectures to find the best structural fit for 1D spectral data:

- **CNN (1D):** Standard convolutional autoencoder. Uses local kernels to detect and "smooth" peak shapes.
- **UNet1D:** Deep encoder-decoder with skip-connections. Excellent for preserving high-resolution peak positions while suppressing noise.
- **Transformer:** Treats spectral channels as a sequence. Uses self-attention to capture long-range dependencies.
- **Conformer:** Combines local convolutions (shape detection) with global attention (peak relationships).

Detailed Model Architectures (2/2)

- **ResNet1D:** Utilizes residual blocks to allow deeper feature extraction without losing original signal information.
- **LSTM:** Recurrent autoencoder. Processes the spectrum as a sequence of energy steps.
- **MLP:** Fully connected autoencoder. Serves as a non-linear baseline.
- **Linear:** Simple linear projection. Provides the ultimate baseline to quantify the value of deep learning.

Loss Function Strategy: Standard MSE + L1

- **MSE:** Ensures global reconstruction of the signal across all 600 channels.
- **L1 Sparsity:** Specifically applied to non-peak regions (where ground truth ≈ 0).
- **Result:** Forces the background to be sparse (near-zero) while maintaining high peak fidelity.

Why this decision?

Standard MSE alone tends to "hallucinate" noise into the baseline. The L1 penalty suppresses these artifacts without affecting peak reconstruction.

Training Stability: Standard Preprocessing

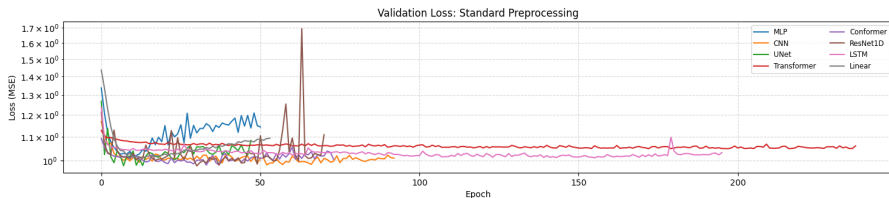


Figure: Validation loss convergence for all models using Standard scaling.

Training Stability: MinMax Preprocessing

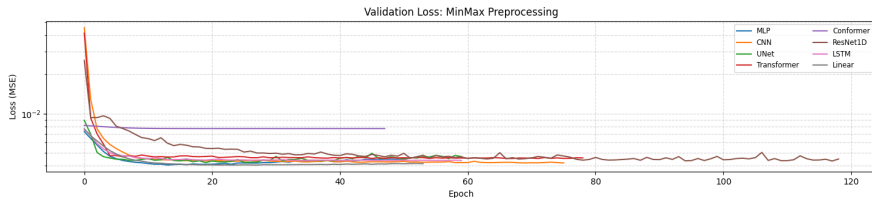


Figure: Validation loss convergence for all models using MinMax scaling.

Training Stability: LogMinMax Preprocessing

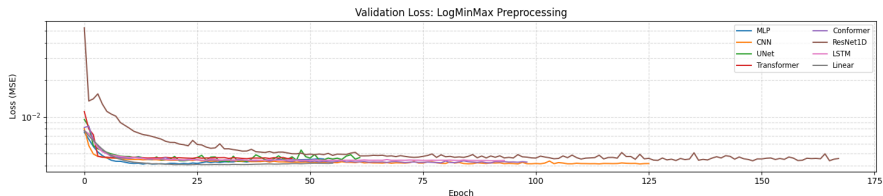


Figure: Validation loss convergence for all models using LogMinMax scaling.

Full Comparative Results: MSE ($\times 10^{-5}$)

Architecture	Standard	LogMinMax	MinMax
CNN	3.58	4.38	4.71
Conformer	3.93	4.45	11.70
LSTM	4.13	4.40	4.24
Linear	4.19	4.77	5.12
Transformer	4.36	4.54	4.55
MLP	4.37	4.92	5.21
UNet1D	4.88	4.63	4.46
ResNet1D	4.94	4.74	4.86

Table: Detailed comparison of MSE across all models and preprocessing strategies.

Full Comparative Results: MAE ($\times 10^{-3}$)

Architecture	Standard	LogMinMax	MinMax
UNet1D	1.20	1.39	1.38
Conformer	1.22	1.30	2.40
CNN	1.25	1.43	1.56
ResNet1D	1.32	1.51	1.56
LSTM	1.34	1.27	1.29
Transformer	1.44	1.40	1.35
Linear	1.66	1.64	1.71
MLP	1.72	1.68	1.70

Table: Detailed comparison of MAE across all models and preprocessing strategies.

Quantitative Evaluation: MAE Grouped Comparison

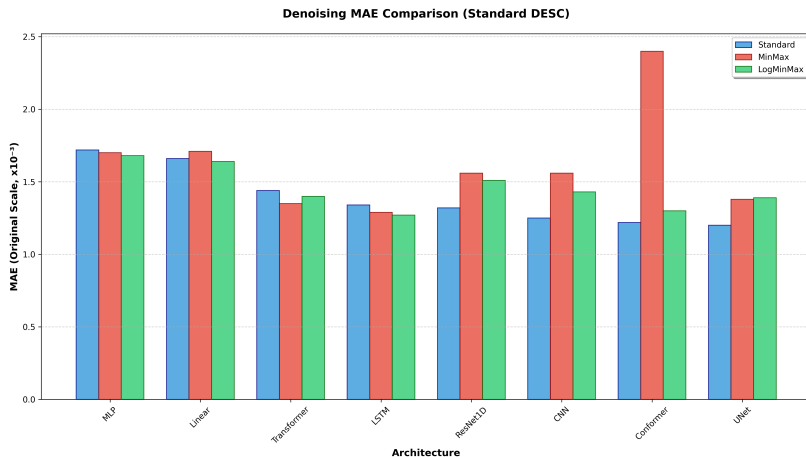


Figure: Impact of preprocessing on architecture performance (MAE Ranking).

Quantitative Evaluation: MSE Grouped Comparison

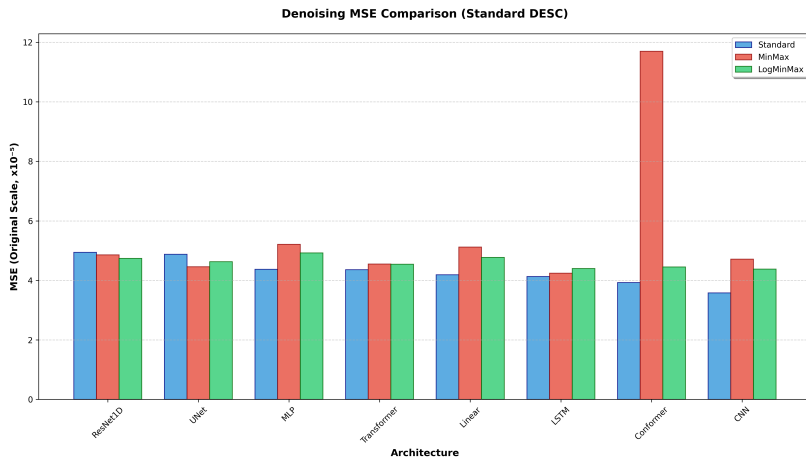


Figure: Impact of preprocessing on architecture performance (MSE Ranking).

Reconstruction Showcase: Best vs Worst Models (MSE)

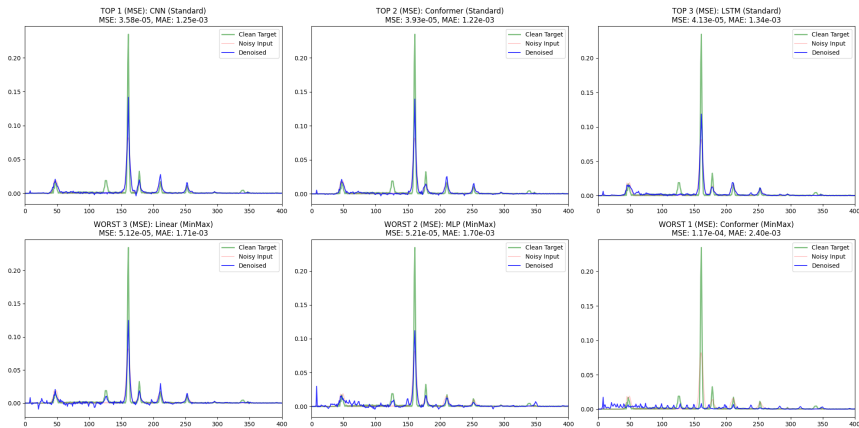


Figure: Visual signal recovery for the overall top 3 and worst 3 models (Standard vs MinMax).

The Denoising Champion: CNN + Standard

- **Global Fidelity:** Achieved the lowest overall MSE across the entire test set.
- **Local Feature Extraction:** The kernels are exceptionally efficient at modeling and recovering peak shapes.
- **Preprocessing Synergy:** Standard scaling preserves the natural peak-to-background ratio which the CNN leverages for optimal noise separation.

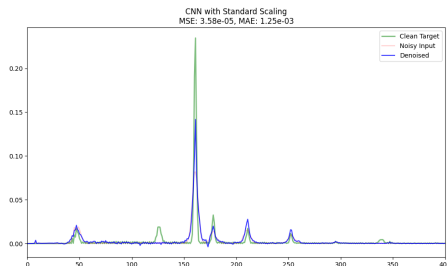


Figure: Superior reconstruction achieved by the CNN Autoencoder.